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SARDAR PATEL UNIVERSITY
B.C.A (SEMESTER-II) EXAMINATION 2026
US02CBCA21 : Advanced C Programming

No. of Printed Pages: 02



Date : 11/04/2026 (Saturday)

Time : 10:00 AM to 01:00 PM

Marks: 70

Q. 1 Multiple Choice Questions

[10]

- 1 The variable declared outside a function is called _____.
(a) global (b) local (c) function (d) none
- 2 Structure is a _____ data type.
(a) built-in (b) derived (c) user defined (d) none
- 3 The parameter is also known as _____.
(a) argument (b) variable (c) data type (d) array
- 4 _____ Operator connects the structure name to its member.
(a) Underscore (b) dot (c) ! (d) none
- 5 Which of the following share a similarity in syntax?
(a) Union (b) Structure (c) Arrays (d) Both a & b
- 6 How will you free the allocated memory?
(a) remove(ptr) (b) free(ptr) (c) dealloc(ptr) (d) destroy(ptr)
- 7 Which of the following defines a pointer variable to an integer?
(a) int &ptr; (b) int **ptr; (c) int &&ptr (d) int *ptr;
- 8 When fopen() is not able to open a file, it returns _____.
(a) EOF (b) NULL (c) one (d) zero
- 9 _____ is datatype of file pointer.
(a) int (b) double (c) string (d) FILE
- 10 The mode _____ is used for opening a file for updating.
(a) r (b) w (c) a (d) r+

Q. 2 Short Question (Any TEN)

[20]

- 1 Write and explain syntax for function call.
- 2 List all the categories of user-defined functions.
- 3 Explain recursion in brief.
- 4 Differentiate structure and union.
- 5 Explain array of structure with example.
- 6 Write and explain syntax for structure variable declaration.
- 7 Explain malloc() with example.
- 8 Define pointer variable.
- 9 Differentiate pointer and pointer variable.
- 10 List basic file operations performed on file.
- 11 Explain fgets() and fputs() with example.
- 12 List out different file modes.

(1)

(P.T.O.)

Q. 3

- [A] Explain what a **function declaration** is. Write its syntax and give an example. [05]
[B] Explain how to pass a **1-D array to a function** with an example. [05]

OR

Q. 3

- [A] Explain a **function with return type and parameters** with an example. [05]
[B] Explain a **recursive function** in simple words with an example. [05]

Q. 4

- [A] What are the different ways to pass a **structure from one function to another**? Explain any one method with an example. [05]
[B] Explain **union** with syntax and example. [05]

OR

Q. 4

- [A] Explain how to **access structure members** with an example. [05]
[B] Explain how memory is shared in a union with example. [05]

Q. 5

- [A] Explain **realloc()** function with an example. [05]
[B] Explain how a **function can return multiple values** with an example. [05]

OR

Q. 5

- [A] Explain **calloc()** function with an example. [05]
[B] Explain **pointer to an array** with an example. [05]

- Q. 6 Explain **fprintf()** and **fscanf()** functions with examples. [10]

OR

- Q. 6 Explain **error handling in file (I/O) operations** with a suitable example. [10]

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